Momin B Khan

Senior AI/ML Engineer | Data Scientist

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Profile

- Senior **AI/ML Engineer** and **Data Scientist** with **12+ years** of experience delivering scalable, production-ready AI solutions across healthcare, finance, and e-commerce — combining advanced analytics, statistical modeling, and modern machine learning engineering
- Skilled in **machine learning**, **NLP**, **LLMs**, and **generative AI**, with hands-on success designing intelligent systems for predictive modeling, vector search, and multimodal data processing
- **Architect** and **deploy end-to-end ML** and **data science pipelines** with deep focus on MLOps, data governance, and explainability across **GCP, AWS, and Azure**, ensuring secure, reproducible, and high-performing AI operations.
- **Trusted** in **high-compliance environments** to translate complex AI capabilities into measurable clinical, financial, and operational impact, driving responsible adoption while upholding privacy and regulatory standards

Education

University Of Virginia,
Bachelor's Degree of Computer Science
2009 – 2013 | Charlottesville, VA

Professional Experience

Elation Health, *Senior AI/ML Engineer | Tech Lead*   Mar 2023 – Present | Remote

Elation AI Insights Platform - *Predictive and Generative Intelligence for Clinical Decision Support*

At Elation Health, I lead the design and deployment of **predictive and generative AI systems** on **GCP**, focusing on patient engagement, risk forecasting, and clinical decision support. I owned the full lifecycle — from data ingestion and model training to scalable deployment and explainable AI integration — ensuring models remain trustworthy, compliant, and impactful in real-world healthcare workflows.

- Led the design and rollout of Elation’s AI Insights Platform, pulling together EHR, clinical, and operational data so care teams could make faster, patient-centered decisions across multiple provider groups.
- Built a **GCP-first ML environment** (**Vertex AI, BigQuery, Dataflow**) that supported both streaming and batch pipelines, which meant near–real-time predictions could flow straight into physician dashboards without downtime.
- Developed large-scale patient risk models in **PySpark/Vertex Pipelines** over just above 10 million EMR records; in practice this lifted early-deterioration and chronic-care prediction accuracy by a little under 30%.
- Set up document-ingestion pipelines with **Document AI, LayoutLMv3,** and **Tesseract** to turn years of scanned and handwritten forms into searchable, structured records — cutting a big chunk of manual data entry for ops teams.
- Worked hand-in-hand with clinicians and data stewards to shift model validation from “nice AUC” to outcome-based metrics like readmission reduction and adherence, so the models actually mapped to clinical value.
- Fine-tuned **Gemini 1.5 Pro** on Vertex AI to summarize notes, discharge instructions, and patient messages; pilots showed review time dropping by roughly a third while keeping medical wording accurate.
- Productionized a **RAG flow** using **LangChain** and **Vertex Matching Engine** so LLM answers stayed grounded in structured EHR/claims data and didn’t leak PHI.
- Used **LoRA-style fine-tuning** and **instruction-tuning** on **Gemini** with Elation’s own clinical corpus to make medication and lab-result summaries more context-aware.
- Ran comparative evals across **Gemini, Claude, GPT-4,** and **Mistral** inside **LangSmith/Promptfoo**, looking at factual consistency, latency, and cost before standardizing on Gemini for production.
- Built a small multi-agent pattern with **LangGraph** and **CrewAI** — one agent extracted entities, another verified facts — which took about 30% of the triage-check workload off nursing staff.
- Added **voice in** and **voice out** (**Whisper + Cloud Text-to-Speech**) so doctors could dictate notes or listen to patient summaries during telehealth visits.
- Exposed model inference through **Go-based gRPC services** on GKE Autopilot with adaptive batching and autoscaling, keeping latency under 100 ms even during clinic peaks.
- Created **PromptOps-style dashboards** to watch token spend, grounding success, and compliance signals so leadership had a clear picture of cost vs. value.
- Reworked **GPU training** for **A100/L4** with **TensorRT**, mixed precision, and CUDA Graphs; training runs finished ~35–40% faster with no quality loss.

- Profiled distributed jobs with **Nsight Systems** and **PyTorch Profiler** to fix kernel and NCCL bottlenecks, pushing GPU utilization past 90%.
- Put GPU-aware batching and load-based scaling on **Cloud Run**, trimming per-inference cost by about 20–25% while keeping latency steady.
- Standardized builds with **NGC CUDA images** and **Terraform** so model environments were reproducible — setup went from "a few days" to "same day."
- Built multimodal feature pipelines in **Apache Beam/Dataflow** to join vitals, labs, notes, and claims into **BigQuery tables**, improving freshness and reliability of the training data.
- Stood up a **Vertex AI Feature Store** with versioning and schema checks so teams could reuse clinical features instead of reinventing them.
- Wired **Great Expectations** and **Evidently AI** into every pipeline to spot schema shifts or distribution drift early and trigger retraining when needed.
- Ran **Fairlearn/Aequitas audits** before production to make sure predictive performance stayed consistent across gender, age, and socioeconomic groups.
- Standardized **LLM evaluation** with **LangSmith, Promptfoo**, and **Ragas** so grounding and factuality got regression-tested, not judged ad hoc.
- Added **SHAP** and **LIME** explainability directly into clinician dashboards so providers could see why the model made a call — a big trust builder.
- Built a **Streamlit rater app** for clinicians to score LLM outputs on clarity, relevance, and tone; their feedback fed into weekly prompt/model updates.
- Unified **model + infra monitoring** with **Cloud Monitoring, Logging, and Trace**, which made finding performance regressions much faster.
- Automated **CI/CD** through **Cloud Build** and **ArgoCD**, including security scans and rollbacks, which cut deployment friction by more than half.
- Connected **MLflow lineage** with **BigQuery audit logs** to get full traceability on datasets, experiments, and model artifacts for HIPAA/FedRAMP reviews.
- Co-led LLM red-teaming with security/compliance to uncover prompt-injection and hallucination issues, and put guardrails in before go-live.
- Worked with **data governance** to lock down encryption-at-rest (KMS), role-segregated IAM, and service isolation so the platform stayed audit-ready.
- Built **cost-telemetry views (GPU hours, latency, token use)** so finance and engineering could forecast compute spend together.
- Created **internal benchmarking suites** to compare Gemini variants and retrievers on cost, latency, and grounding for future-scale decisions.
- Rolled out **A/B tests** for model-driven interventions and tracked how they changed provider engagement and clinical outcomes over several quarters.
- Wrote **Python utilities** for **GPU benchmarking, RAG traceability,** and **prompt versioning** that later became shared internal libraries.
- Mentored teammates on LLMOps, prompt safety, GPU profiling, and reproducible ML to raise the team's floor.
- Partnered with clinical PMs and designers to bring model outputs into **React dashboards** providers already used — no extra tool to learn.
- Consolidated older prediction services into a single GCP MLOps framework, which made releases faster and audits simpler.
- Presented Elation's **Responsible AI playbook** to execs and partners and helped align the AI roadmap with population health, operations, and value-based care goals.

Cognizant, *Senior Data Scientist / Machine Learning Engineer* Mar 2019 – Feb 2023 | Remote

***OmniAI Delivery Suite** – Multi-Cloud Machine Learning and AI Consulting for Regulated Industries*

At Cognizant, I delivered **data-driven solutions** across healthcare, finance, and e-commerce clients by combining **statistical modeling, machine learning, and business analytics**. I translated ambiguous client objectives into measurable analytical strategies, design interpretable models, and implement **multi-cloud ML workflows** that balance rigor, scalability, and regulatory compliance.

- Led data science engagements from exploration to production, turning messy client data into decisions clinicians, marketers, or finance leaders could actually act on.
- Designed and validated predictive models in **Python, R,** and **MATLAB**, picking the approach (statistical vs ML) based on the client's data quality, volume, and regulatory constraints.

- For healthcare clients, ran multivariate regression, time-series decomposition, and factor analysis in MATLAB to forecast admissions and plan hospital staffing and beds.
- Applied **Bayesian hierarchical models** to cost-of-care estimation so actuaries could talk about *uncertainty* and not just point estimates — better pricing for insurance portfolios.
- Built customer/audience segments with k-means, GMMs, and hierarchical clustering to surface high-value groups and the levers that kept them from churning.
- Created credit-risk scoring pipelines using **logistic regression, decision trees,** and **LightGBM**, which brought loan default misclassification down by roughly 19% for one financial client.
- Developed **demand-forecasting** and **price-elasticity models** for retail (**ARIMA, Prophet, regression** with external signals) so promo and inventory teams could plan more confidently.
- Did the heavy feature-engineering work on large transaction datasets, pulling out 40+ stable predictors that became the default feature set for multiple downstream models.
- Ran **A/B** and **multivariate tests** with **Bayesian/sequential approaches** so marketing teams could stop tests early without losing statistical power.
- Built **Monte Carlo simulation frameworks** to model uncertainty in patient-cost prediction and operational throughput when inputs were moving targets.
- Shipped **forecasting and KPI dashboards** in **Tableau** and **Power BI**, wiring in R/Python scripts so model outputs refreshed for execs without manual exports.
- Used survival / time-to-event analysis to spot readmission-prone patients and give clinical teams a list to follow up on.
- Ran causal impact and uplift modeling to separate "this happened" from "this happened *because of the campaign*."
- Built **text-analytics pipelines (TF–IDF, spaCy, early transformer models)** to summarize physician notes and call-center logs for sentiment and compliance flags.
- Deployed the final models to production on **AWS SageMaker** and **Azure ML** for several healthcare and retail clients, wrapping training and inference in containers so we could scale cleanly and still meet their security/compliance requirements.
- Wrapped **MATLAB econometric models** in **Python APIs** so analytics prototypes could actually be deployed in client environments.
- Used PCA and discriminant analysis to shrink noisy diagnostic feature sets down to something clinicians could read and trust.
- Set up **forecast validation** with backtesting, rolling-origin evaluation, and residual checks so models didn't fall apart the second seasonality shifted.
- Partnered with finance to blend statistical forecasts with what-if trees in R to look at risk-adjusted profitability, not just topline.
- Built **sampling / resampling (bootstrap, jackknife) pipelines** for healthcare and claims data to get reliable variance and confidence intervals.
- Trained and mentored client teams on experiment design, regression diagnostics, and — most importantly — how to explain uncertainty to non-technical stakeholders.
- Worked with PMs to turn stats output into narratives and recommendations, not just notebooks.
- Added automated data validation in Python (Great Expectations) to flag drift, outliers, or missing fields in live feeds before they hit the model.
- Ran **sensitivity analysis** on financial models to show which inputs actually moved the business outcome.
- Built hierarchical time-series models for multi-region healthcare ops so budgeting teams could allocate with more precision.
- Created **interactive R Shiny views** that showed intervals and model drift over time — good for execs who don't want to read code.
- Helped data engineers migrate **legacy SAS workloads** into **PySpark/R pipelines**, cutting processing time by about 70%.
- Standardized experiment tracking in **MLflow** and **RMarkdown** so every dataset, run, and parameter had lineage for audits.
- Delivered client-facing insight reports in **RMarkdown/LaTeX**, mapped to the KPIs leadership cared about.
- Contributed playbooks and case studies to Cognizant's AI/analytics CoE so newer analysts could reuse pipelines instead of starting from scratch.
- Presented results in workshops with client leadership, combining visual storytelling with the stats underneath to get projects actually adopted.

Wayfair, *Data Scientist* Jan 2016 – Feb 2019 | Boston, MA

NovaRec Platform *– Intelligent Product Recommendations and Search Optimization for E-Commerce Personalization*

At Wayfair, I developed and productionize **data science solutions** for personalization, search relevance, and pricing optimization. I owned end-to-end experimentation — from data exploration and model design to **A/B testing, inference, and reporting** — to drive measurable improvements in customer experience and business performance across the e-commerce platform.

- Joined Wayfair's central data science group to build large-scale personalization systems that influenced ranking, on-site experience, and marketing decisions for millions of daily sessions.
- Built both **collaborative-filtering** and **content-based recommenders** in **Python** and **Scala** on **Spark MLlib**; across several categories we saw product CTR and conversion lift by a bit over 12%.
- Designed **search-relevance models** that combined learning-to-rank (**LightGBM**) with embedding-based semantic search, which helped long-tail queries surface better results and improved NDCG by about 15%.
- Ran **A/B tests** and **causal uplift analyses** to measure recommendation impact; introduced Bayesian/sequential testing so teams could stop experiments earlier without losing statistical strength.
- Developed pricing and discount-optimization models in R using regression and elasticity analysis, enabling dynamic promo strategies that lifted gross margin per session by roughly 6%.
- Built shared feature pipelines in **PySpark** and **Airflow** to standardize behavioral, inventory, and session-level signals for all downstream ML/analytics teams.
- Shipped real-time recommendation endpoints using **AWS Lambda** with **DynamoDB-backed APIs** so pages could personalize on render, not on refresh.
- Worked with front-end engineers to add **React-based experimentation dashboards**, giving merchandising and growth teams a live view of model impact.
- Used **text and image embeddings** (**TensorFlow/Keras**) to improve product-similarity scoring, connecting structured catalog data with unstructured titles, descriptions, and imagery.
- Wrote internal, data-informed guidance for PMs, UX, and merchandising on how to read A/B results and when to lean on predictive signals for ranking or placement decisions.

Capital One, *Data Analyst / Data Science Associate* Jul 2013 – Dec 2015 | McLean, WA

Risk Analytics Modernization *– Predictive Modeling and Business Intelligence for Consumer Credit Operations*

At Capital One, I supported **credit risk, fraud detection, and financial forecasting** through statistical analysis, data automation, and predictive modeling. I collaborated with risk strategy teams to transform raw transaction and behavioral data into actionable insights that improve **approval precision, portfolio performance, and compliance readiness**.

- Started in the Risk Analytics group as a data analyst and grew into a data science associate role, supporting credit, fraud, and marketing lines with modeling and automation.
- Built **credit-risk scoring models** in **Python** and **SAS (logistic regression, decision trees)** to tighten approval precision while staying within portfolio risk appetite.
- Ran **EDA** and **hypothesis tests** on millions of card and deposit transactions to spot behavioral patterns that later fed into underwriting rules.
- Developed **fraud/anomaly detection jobs** on **Hadoop/Hive** using time-window aggregations and outlier scoring to flag suspicious account activity early.
- Automated data prep and feature extraction in Python/SQL, which cut model refresh time for quarterly updates from days down to hours.
- Built **AngularJS dashboards** on top of Java-based APIs so risk managers could see portfolio health, roll rates, and default trends in something close to real time.
- Put together forecasting frameworks in R and MATLAB to project delinquency and loss trends, helping finance with long-term capital planning.
- Worked with data engineering to move early analytics workloads onto **AWS (Redshift, EMR),** which was my first exposure to distributed processing in production.
- Designed loss-forecasting and credit-limit recommendation models using regularization and simple ensembles to improve generalization on out-of-time samples.
- Presented results to senior risk leaders focusing on interpretability — lift charts, KS, segmentation views — so model changes could be approved and rolled into strategy

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Skills

Languages & Frameworks
Python, Go, R, MATLAB, JavaScript, TypeScript, SQL, Bash, Flask, FastAPI, Django, Jupyter, PySpark, Scala, Node.js, Express, React, Angular

NLP & Language Modeling
spaCy, BERT, RoBERTa, GPT-Neo, Sentence Transformers, TF-IDF, BM25, Named Entity Recognition (NER), Text Classification, Topic Modeling, Document Clustering, Semantic Similarity, LayoutLMv3, Tesseract OCR, Whisper (Speech-to-Text), Polly (Text-to-Speech), Gemini Pro, Vertex AI PaLM APIs, LangGraph, Semantic Kernel, DSPy

Vector Search & Retrieval
FAISS, Pinecone, Weaviate, Chroma, BM25, Hybrid Dense–Sparse Retrieval, Embedding Pipelines, LangChain Retrieval, Unstructured.io, GraphRAG, Vertex Matching Engine

MLOps & Model Lifecycle
Vertex AI, AI Platform (GCP), SageMaker, Azure ML, Databricks, MLflow, Weights & Biases, Airflow, Docker, Kubernetes (GKE, AKS, EKS), Triton Inference Server, ONNX Runtime, ZenML, Feast (Feature Store), Modelbit, MLServer, Kubeflow, TensorRT Serving, Vertex AI Pipelines

Cloud & DevOps Platforms
AWS (S3, Glue, Lambda, ECS, Redshift, DynamoDB, Step Functions, Bedrock, SageMaker)
GCP (BigQuery, Dataflow, Pub/Sub, Cloud Functions, Cloud Run, Cloud Storage, Vertex AI, Looker Studio, Matching Engine)
Azure (Blob Storage, Event Hubs, Cosmos DB, Azure Databricks, Azure ML, Key Vault, Azure DevOps
Terraform, GitHub Actions, Cloud Build, Jenkins, Helm, ArgoCD)

Databases & Storage
BigQuery, PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Cosmos DB, Redshift, Athena, S3, Data Lake Storage, Parquet

Machine Learning & Deep Learning
scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Keras, Transformers, CNNs, RNNs, Attention Models, GANs, LoRA, ONNX, FlashAttention, DeepSpeed, xFormers

LLMs, Generative AI & Agent Frameworks
Gemini, GPT-4, Claude, FLAN-T5, LLaMA 2, Instruction Tuning, RLHF, Few-Shot Learning, Prompt Engineering, Prompt Chaining, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, CrewAI, AutoGen, Langflow, LangSmith, Ragas, Promptfoo, Human-in-the-Loop Evaluation, Hallucination Mitigation, Guardrails AI

GPU & Accelerator Computing
NVIDIA CUDA, cuDNN, TensorRT, NCCL, GPU-optimized TensorFlow/PyTorch, Google TPU, Vertex AI Training on A100/L4 instances

Monitoring, Testing & Explainability
Prometheus, Grafana, Cloud Logging, Cloud Monitoring, Great Expectations, Evidently AI, Drift Detection, SHAP, LIME, TruLens, GPT-as-a-Judge, Ragas, Fairlearn, Aequitas, Counterfactual Explainers, LLM Red-Teaming, Bias & Safety Auditing, PHI-Safe Evaluation

Visualization & Analytics Tools
Streamlit, Tableau, Power BI, Looker Studio, Plotly, Matplotlib, Seaborn, QuickSight, R Shiny, Dash, JupyterLab, LaTeX, RMarkdown

Statistical Analysis & Experimentation
MATLAB (Statistics, Econometrics, Optimization, Signal Processing), R (tidyverse, ggplot2, forecast, causalImpact), Bayesian Inference, Time-Series Analysis, Monte Carlo Simulation, A/B Testing, Causal Modeling, Uplift Analysis